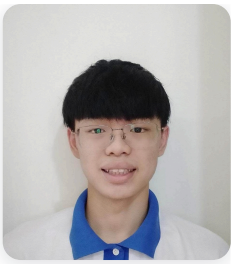
MAH QING TONG

LLM ENGINEER | AI ENGINEER

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PROFESSIONAL SUMMARY

LLM Engineer at Zetrix AI with a Bachelor of Computer Science in Data Science from **Monash University Malaysia**. Passionate about exploring autonomous agent architectures, production Agentic RAG workflows, and modular **Agent Skills (`SKILL.md` / MCP)** with practical harness engineering and systematic evaluation. Frequently serving as the Person-in-Charge (PIC) across key AI initiatives, demonstrating high ownership, technical accountability, and end-to-end responsibility from initial system design to production delivery. Proven expertise in hybrid dense/sparse retrieval, cross-encoder reranking, and full-stack backend development for robust enterprise AI applications.

TECHNICAL SKILLS

LLM & Agentic Systems:	LangChain, LangGraph, Agent Skills (SKILL.md), Model Context Protocol (MCP), Dify, Multi-Agent Orchestration, Human-in-the-Loop (HITL), Prompt Engineering
RAG & Information Retrieval:	Hybrid Search, Dense & Sparse Retrieval (BM25), Qdrant, ChromaDB, Cross-Encoder Reranking, Multi-Hop Retrieval, Hierarchical Ingestion
Programming & Backend:	Python, FastAPI, TypeScript, JavaScript, SQL, REST APIs, Next.js, Server-Sent Events (SSE)
Machine Learning:	PyTorch, Hugging Face Transformers, BERT Prefix Prompt-Tuning, CNNs, Embeddings, Self-Attention from scratch
Databases & DevOps:	PostgreSQL, DuckDB, MongoDB, Git, Docker, Linux, Pytest, GitHub Actions, CI/CD

PROFESSIONAL EXPERIENCE

Zetrix AI | LLM Engineer — JUN 2026 – PRESENT

- **Enterprise AI Workflow Automation:** Developed Dify-based AI workflows for enterprise SOP and knowledge-base retrieval, grounding LLM responses in internal documentation to improve answer relevance and consistency for customer operations.
- **Section-Aware RAG Ingestion:** Built automated preprocessing pipelines to convert unstructured PDF research into hierarchical Markdown before chunking, improving downstream retrieval recall by 28% in measured evaluations.
- **Agentic RAG & Deterministic Execution:** Developed the i-FIKR Islamic knowledge agent over 870+ records, integrating vector search with custom Python and Pandas tools for accurate multi-criteria filtering and tally operations.
- **Multi-Hop Retrieval & Citation Grounding:** Implemented a two-stage retrieval workflow combining catalog-level metadata grounding with targeted document retrieval, enabling cross-document comparisons with verifiable sub-chunk citations.
- **Conversational Booking Agent:** Built an interactive property booking agent supporting real-time availability checks, appointment booking, schedule reviews, and Human-in-the-Loop (HITL) handoffs.

NIQ (NielsenIQ) | Data Input Validation Intern — NOV 2024 – FEB 2025

- **Large-Scale Dataset Validation:** Processed and validated retail datasets containing thousands of records per batch, ensuring data consistency and integrity for downstream analytics.
- **Rule-Based Anomaly Detection:** Applied logic-driven validation techniques to identify catalog inconsistencies and data anomalies, improving regional data reliability.
- **Cross-Regional Collaboration:** Collaborated with Malaysia and Vietnam teams to investigate data issues and standardise data validation practices.

TECHNICAL PROJECTS

AI application development • RAG • Agentic systems • Backend engineering

Personal DeepAgent — AI Agent System & Kinetic Portfolio `LangGraph` • `Agent Skills` • `Hybrid RAG` • `FastAPI` • `DuckDB` • `FlashRank` • `Next.js` • `SSE`

- Architected a multi-agent orchestration graph featuring dynamic planning with step-budget guardrails, a ReAct research agent, and modular sub-agent workers with isolated execution.
- Implemented modular Agent Skills (`SKILL.md` / MCP) and hybrid retrieval combining dense vector embeddings with sparse BM25 indexing in DuckDB and FlashRank cross-encoder reranking.
- Engineered deterministic tool execution for system commands, multi-engine web search, and file operations with automated failure recovery and self-correction loops.
- Integrated a full-stack kinetic portfolio web interface (Next.js, Tailwind CSS) powered by an asynchronous FastAPI backend with Server-Sent Events (SSE) streaming and Human-in-the-Loop approval workflows.

AI Job Search & Resume Agent `LangChain` • `MCP` • `Qdrant` • `Hybrid Search` • `JSearch API`

- Designed a multi-agent LLM application automating job discovery, semantic candidate ranking, resume analysis, and personalised email drafting.
- Implemented dense and sparse BM25 hybrid retrieval with cross-encoder reranking in Qdrant for high-precision resume-to-job matching.
- Developed an orchestration workflow managing query rewriting, information retrieval, ranking, and final response generation.

Smart Expense Tracker by OCR & AI Analytics `Kotlin` • `Firebase` • `Taggun OCR` • `Real-Time Database` • `Mobile Architecture`

- Developed an end-to-end mobile application using Kotlin and Firebase, enabling real-time database synchronization and secure user authentication.
- Integrated Taggun OCR and AI-driven categorization to automate receipt text extraction and generate real-time financial insights for users.

EDUCATION

Monash University Malaysia | **Bachelor of Computer Science in Data Science** FEB 2023 – DEC 2025

Second Upper Honours • CGPA: 3.36 / 4.00

- **Deep Learning & Neural Architectures:** Implemented Multi-Head Self-Attention modules, Position-wise Feed-Forward networks, and Transformer Encoders from scratch in PyTorch; architected Prefix Prompt-Tuning pipelines for BERT adaptation and evaluated RNN/LSTM/GRU sequential models across multi-class benchmarks.
- **Computer Science & Data Science Foundations:** Mastered advanced Algorithms & Data Structures, Big Data Management, Data Visualization, Relational & Distributed Databases, Cybersecurity principles, and Object-Oriented Software Engineering.

LEADERSHIP & HONORS

- **President of Diabolo Society:** Foon Yew High School (Jan 2019 – Dec 2021) — Directed society operations, organized performances, led weekly training programs, and coordinated major cultural events.
- **Monash Coding League 2024:** Competitive algorithmic programming challenge.
- **ExxonMobil DataWorks Challenge 2024:** Data analytics and predictive modeling competition.
- **Leadership Champ by YCA:** Youth leadership accreditation.